

Market Insurance Project — Problem Identification

Harvesting the Volatility Risk Premium with Python (yfinance)

1. Problem Statement

Equity index options are systematically overpriced relative to the volatility that is subsequently realized in the underlying market. This gap exists because option buyers are willing to pay an insurance-like premium to hedge against sharp downside moves, while option sellers demand compensation for bearing that tail risk. The result is that implied volatility (IV), the market's forward-looking estimate of price fluctuation embedded in option prices, tends to exceed realized volatility (RV), the volatility that actually occurs over the life of the contract, more often than not.

This project investigates whether a systematic, rules-based options-selling strategy can capture this premium and generate a return stream that is both consistent over time and largely uncorrelated with traditional long-only equity returns. The problem sits squarely within the scope of the course: it requires collecting and cleaning real financial time-series and options-chain data, engineering volatility features, and using exploratory and statistical analysis to test a well-defined hypothesis, all core Data Science Methodology (DSM) skills.

2. Problem Identification — DSM Substeps

2.1 Background and Business Context

Institutional and retail traders alike search for return streams that do not simply track the stock market (i.e., have low correlation to the S&P 500). Selling options is a well-documented source of such a return stream, but it is also a strategy that can suffer large, infrequent losses if not managed with data-driven risk rules. Understanding when and how the VRP is most reliably positive is therefore a genuine business/investment problem, not just an academic exercise.

2.2 The Business/Analytic Question

- Primary question: Is implied volatility on major equity index and ETF options (e.g., SPY) persistently higher than the volatility subsequently realized, and can this spread be harvested as a systematic, uncorrelated income strategy?
- Supporting questions: How large and how stable is the IV–RV spread over time? Does it vary predictably with market regime (e.g., VIX level, time to expiration)? What is the correlation of a simple short-volatility strategy's returns to SPY's returns?

2.3 Client / Stakeholder Perspective

The intended "client" is an individual or small systematic trading desk evaluating whether a volatility-selling overlay belongs in a diversified portfolio. The stakeholder needs evidence-based answers (not intuition) on premium persistence, drawdown risk, and correlation benefits before allocating capital.

2.4 Scope of the Problem

The project is scoped to liquid U.S. equity index proxies (SPY and/or QQQ) and their listed options, using near-term expirations (roughly 7–45 days to expiration) where the VRP effect is most studied and most liquid. It excludes exotic derivatives, single-name equity options with low liquidity, and live/automated trade execution — the focus is on identification, measurement, and validation of the opportunity using historical data, which fits the course's data science scope rather than a live-trading engineering scope.

2.5 Data Requirements

To test the problem statement, the project requires: (a) historical daily price data for the underlying (to compute realized volatility), (b) historical or point-in-time options-chain data with strike, expiration, and implied volatility fields (to compute implied volatility and simulate a short-options position), and (c) a market volatility benchmark (the VIX index) to contextualize regimes and correlation.

2.6 Data Collection Plan (Dataset Selection)

The Python yfinance library provides a single, free, and programmatically accessible source that satisfies all three data requirements above, which is why it was selected:

yfinance Method	Data Retrieved	Role in the Problem Statement
Ticker.history()	Daily OHLC price history for SPY / QQQ	Used to compute rolling realized (historical) volatility, the benchmark against which implied volatility is compared.
Ticker.option_chain()	Strikes, expirations, bid/ask, and implied volatility for listed options	Used to extract implied volatility at target strikes/expirations and to price the hypothetical short-option ("insurance-selling") positions.
Ticker.history() on ^VIX	Daily VIX index level	Used as a market-wide implied-volatility benchmark and to segment the analysis by volatility regime.

2.7 Feasibility and Success Criteria

- Feasibility: yfinance is free, requires no API key, and returns data in a pandas-ready format, making rapid collection and iteration realistic within course timelines.
- Success criterion 1: A measurable, positive average spread between implied and realized volatility over the sample period.
- Success criterion 2: A simulated short-volatility return series with correlation to SPY buy-and-hold returns that is meaningfully below 1.0, supporting the "uncorrelated income" claim.
- Success criterion 3: Clear identification of the conditions (e.g., VIX regime, days-to-expiration) under which the premium is most and least reliable, to inform later modeling steps.

3. Summary

The problem, quantifying and systematically harvesting the volatility risk premium, is well-defined, testable, and appropriately scoped for this course. The yfinance library supplies all the underlying price, options, and volatility-index data needed to move into the next DSM stages (data understanding, preparation, and analysis) without requiring paid data subscriptions or infeasible data engineering.